

NISHA KARKI

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Professional Summary

- Senior Data Engineer with extensive experience in AI-driven financial analytics, large-scale data engineering, and cloud-based infrastructure development, specializing in machine learning, Apache Spark, Kafka, and AWS to optimize data pipelines, risk management systems, and predictive modeling in financial and healthcare sectors.
- Expertise in AI and machine learning, developing AI-driven trading analytics, portfolio optimization, and fraud detection systems using TensorFlow, MLflow, and Amazon SageMaker. Built deep learning-based predictive models to analyze historical market trends, optimize asset allocation, and detect anomalies in financial transactions.
- Strong background in data engineering and real-time analytics, leveraging Apache Spark, PySpark, Databricks, and Kafka Streams to design and implement scalable ETL pipelines for structured and unstructured datasets. Engineered high-throughput data processing solutions for financial transactions, market data streams, and healthcare records, improving decision-making efficiency.
- Cloud computing expertise across AWS and Azure, deploying microservices-based financial and healthcare solutions using Amazon EKS, AWS Lambda, Amazon S3, Amazon RDS, and Azure Functions. Implemented scalable, high-availability architectures for real-time AI inference, investment analytics, and telehealth services.
- Proficient in SQL optimization and database management, working extensively with Snowflake, PostgreSQL, MySQL, and Amazon RDS. Designed partitioned, indexed, and clustered data storage solutions to enhance query performance, ensuring rapid retrieval of financial, investment, and healthcare records.
- Experienced in workflow orchestration, automation, and data governance, leveraging Apache Airflow to schedule ETL jobs, manage AI model training pipelines, and enforce data quality controls. Implemented governance policies for data lineage tracking, access control, and regulatory compliance in financial risk assessment workflows, ensuring secure and auditable data processing.
- Strong API development skills with RESTful architectures, designing secure and scalable APIs using FastAPI, Django REST Framework (DRF). Integrated OAuth 2.0 authentication and RBAC policies to ensure data security in financial, healthcare, and trading applications.
- Full-stack development expertise, building React.js-based investment dashboards and real-time monitoring systems for financial and healthcare analytics. Developed dynamic frontends for market trend visualization, risk analysis, and AI-powered trade execution monitoring, integrating with backend microservices.
- Advanced experience in real-time streaming and messaging architectures, implementing Apache Kafka, WebSockets, and RabbitMQ to enable instant updates for trading platforms, telehealth services, and financial risk models. Built event-driven systems to handle high-velocity data processing with fault tolerance and scalability.
- Proven ability to enhance system performance and reduce latency, leveraging Redis caching, Celery for background processing, and Kubernetes for efficient container orchestration. Optimized high-load applications for financial transactions, investment analytics, and AI model inference.
- Expert in continuous integration and deployment (CI/CD), automating build, testing, and deployment workflows using Jenkins, Azure DevOps, and GitHub Actions. Ensured seamless software updates for AI-driven financial and healthcare platforms, maintaining high availability and security compliance.
- Proficient in Power BI, designing interactive financial dashboards for portfolio risk analysis, trade execution monitoring, and investment performance tracking. Integrated real-time data pipelines to provide actionable insights to traders and financial analysts, enabling data-driven decision-making.

Experience

Senior Data Engineer

Charles Schwab

Aug 2021 - Present

- Developed an AI-driven trading analytics platform to optimize portfolio risk management and asset allocation. The system used machine learning models to predict market trends, analyze historical trading patterns, and detect anomalies in financial transactions. This solution enhanced real-time investment strategies, improved risk assessment, and optimized trading execution efficiency.
- Designed and implemented AI-driven data pipelines using Apache Spark, PySpark, and Snowflake, enabling seamless ingestion, transformation, and warehousing of structured and unstructured market data. Leveraged Snowflake's scalable cloud data warehouse to centralize real-time stock trades, financial transactions, and behavioral analytics, improving data accessibility for downstream ML models and enabling faster, compliance-ready analytics across portfolio risk and market prediction use cases.
- Automated financial model orchestration using Apache Airflow, scheduling ETL workflows, machine learning model training, and market data aggregation processes. Implemented data governance policies to ensure data lineage tracking, access controls, and compliance with financial regulations, maintaining data integrity and security for AI-driven risk assessments.

- Developed deep learning-based portfolio optimization models with TensorFlow, analyzing historical trade data, market volatility, and macroeconomic indicators to recommend optimal asset distributions. By leveraging neural networks, these models provided risk-adjusted investment recommendations, helping traders and financial advisors make informed decisions based on complex, non-linear financial data patterns.
- Engineered real-time AI inference pipelines using Kafka Streams and AWS Lambda, ensuring instant updates to portfolio risk assessments and trade execution strategies. Kafka Streams enabled continuous processing of live market data, while AWS Lambda executed lightweight inference tasks on the fly, allowing real-time adjustments to investment strategies without infrastructure overhead.
- Optimized machine learning model deployment using Databricks and Amazon SageMaker, leveraging Databricks for collaborative notebooks and accelerated ML pipelines on Apache Spark. SageMaker handled model training, hyperparameter tuning, and deployment, while Databricks improved collaboration and iteration speed on data preprocessing, feature engineering, and model validation, leading to faster insights and more accurate financial forecasting.
- Automated financial model orchestration using Apache Airflow, scheduling ETL workflows, machine learning model training, and market data aggregation processes. Apache Airflow reduced manual intervention in AI pipeline execution, ensuring that models were consistently trained on up-to-date financial data, improving the accuracy of trading signals and risk assessments.
- Developed an advanced trade execution analytics platform, leveraging AI and machine learning to monitor order flow, detect price slippage, and optimize trade execution strategies. By processing real-time transaction data, this system improved trade efficiency, reduced execution costs, and enhanced market liquidity analysis for institutional investors.
- Deployed AI-powered microservices on AWS, utilizing Amazon EKS for Kubernetes-based container orchestration, AWS Lambda for serverless execution, and Amazon S3 for scalable financial data storage. This cloud-based infrastructure provided high availability, fault tolerance, and efficient management of machine learning models used for financial forecasting and risk analysis.
- Designed and developed interactive Power BI dashboards for portfolio risk analysis and investment performance tracking. Integrated real-time financial data pipelines to provide traders and financial analysts with dynamic visual insights into market trends, trade execution efficiency, and risk exposure.

Data

American Express

Engineer

May 2019 - Aug 2021

- Designed and implemented scalable ETL pipelines using PySpark on Databricks, leveraging its optimized Spark environment to process vast biomedical datasets. This reduced data processing time significantly and ensured efficient integration of both structured and unstructured patient records for advanced analytics and compliance reporting.
- Developed and optimized complex SQL queries in Snowflake, leveraging indexing, partitioning, and clustering strategies to enhance query performance, reduce execution time, and efficiently retrieve large-scale clinical and pharmaceutical datasets for research and regulatory compliance teams.
- Built a data lake architecture on AWS S3, enabling secure storage of high-volume genomic, clinical, and real-world healthcare data while ensuring seamless integration with analytics tools, improving accessibility for data scientists and facilitating large-scale AI/ML-driven medical research.
- Engineered batch and streaming data ingestion pipelines using Apache Kafka, ensuring reliable, real-time data integration from electronic health records (EHRs), IoT-enabled medical devices, and lab systems, improving decision-making speed and enabling near-instant updates for clinical trial monitoring.
- Automated data workflow orchestration using Apache Airflow, optimizing ETL job dependencies, scheduling batch jobs efficiently, and ensuring timely data availability for AI-driven analytics and predictive modeling in pharmaceutical research and drug development.
- Developed RESTful APIs using FastAPI to enable secure and scalable data access across research teams, integrating OAuth 2.0 authentication and RBAC, ensuring data privacy compliance while facilitating seamless communication between applications and clinical databases.
- Implemented data validation and integrity checks using Great Expectations, automating schema validation, anomaly detection, and missing data handling, ensuring high-quality, standardized datasets for clinical research, patient insights, and FDA compliance reporting.
- Designed and built AI-driven data pipelines using TensorFlow, integrating large-scale patient records and biomedical datasets to develop predictive models for disease progression, patient risk stratification, and personalized treatment recommendations, enhancing decision-making for clinical researchers.
- Optimized machine learning workflows by creating efficient feature engineering pipelines in PySpark, automating data preprocessing, missing value imputation, and outlier detection, significantly reducing model training time and improving predictive accuracy for healthcare-related AI applications.

Python Engineer

T. Rowe Price

Jan 2018 - May 2019

- Developed and maintained a Python-based Django backend system to process high-volume investment transactions, ensuring seamless interaction with PostgreSQL via Django ORM. Implemented structured query optimization techniques to enhance financial data retrieval, improve transaction processing speeds, and support compliance.
- Designed and implemented RESTful APIs using Python and Django REST Framework (DRF) to enable secure access to investment performance data, risk analytics, and trade execution insights. Used serializers for data validation and error handling, ensuring seamless integration with trading platforms, portfolio management tools, and regulatory systems.
- Built an interactive investment risk analytics dashboard using React.js, HTML, and CSS, providing real-time visualization of market trends, portfolio allocations, and risk exposure. Integrated dynamic state management in React to enable live updates on asset performance, enhancing decision-making for portfolio managers and investment analysts.
- Optimized financial data storage and retrieval in PostgreSQL by implementing indexing, partitioning, and stored procedures. This significantly improved query performance, allowing traders and analysts to quickly access and analyze historical transaction data for investment trend forecasting.
- Integrated Redis caching to optimize API response times for retrieving frequently accessed portfolio risk scores and historical market data. This reduced direct database queries and improved real-time insights for investment advisors managing large client portfolios.
- Automated ETL workflows using Python and Apache Airflow to process and ingest daily market data from multiple financial sources. This ensured that investment analytics models and trading algorithms had up-to-date market intelligence for asset allocation and risk assessment.
- Developed a real-time trade monitoring and alerting system using WebSockets, enabling instant notifications for market fluctuations and regulatory compliance violations. This allowed traders to react to significant price movements and execute timely investment decisions.
- Configured Celery with RabbitMQ to offload intensive trade reconciliation tasks to background workers. This ensured high availability of investment data APIs, preventing performance bottlenecks even during peak market trading hours.
- Deployed scalable financial data processing microservices on AWS, utilizing Amazon EKS for Kubernetes container orchestration, AWS Lambda for automated trading reports, Amazon RDS (PostgreSQL) for structured trade data storage, and Amazon S3 for archival of historical financial statements and investment performance reports.

Python Developer

UnitedHealth Group

Jan 2017 - Dec 2017

- Developed a microservices-based telehealth solution using Python and FastAPI, enabling secure and scalable healthcare service integration. Leveraged REST APIs to facilitate seamless communication between services, ensuring efficient data exchange while maintaining compliance with healthcare regulations.
- Designed and optimized SQL queries and stored procedures to enhance real-time patient data retrieval, integrating with PostgreSQL and MySQL databases. This improved system performance and ensured accurate and fast access to medical records, supporting quick decision-making for healthcare professionals.
- Enhanced interoperability between Python and Ruby-based services by improving API communications and background job processing, particularly for automated appointment reminders. Used Python to trigger Ruby scripts, ensuring timely notifications for patients and reducing appointment no-shows.
- Deployed the telehealth platform on Azure, utilizing Virtual Machines for hosting, Azure Functions for event-driven tasks, Blob Storage for secure data archiving, and SQL Database for structured storage. This ensured high availability, scalability, and compliance with healthcare industry standards.
- Leveraged Apache Spark to process large-scale healthcare data, analyzing patient history and medication adherence trends. This facilitated predictive analytics, allowing doctors to make more informed treatment decisions based on historical data patterns.
- Designed and developed secure REST APIs to ensure seamless integration with third-party healthcare systems. Focused on optimizing API endpoints for performance and reliability, enabling smooth interoperability with external healthcare providers, insurance systems, and electronic health record (EHR) platforms.
- Integrated Kafka for real-time event streaming, allowing secure, instant data flow between services. Additionally, used WebSockets with RBAC policies for encrypted real-time messaging and video conferencing, ensuring private and secure telehealth sessions between patients and providers.
- Automated deployments and software updates by implementing CI/CD pipelines using Jenkins and Azure DevOps. This streamlined development workflows, reducing manual effort and ensuring continuous integration and deployment of new features with minimal downtime.
- Managed containerized microservices using Docker for consistency and Kubernetes for orchestration, enabling seamless scaling of healthcare services. This architecture improved system reliability by efficiently handling varying loads and ensuring high availability of critical services.

Education

- Bachelor of Engineering, RGPV, Indore – 2012